Mayank Shah

+91-7980349041 | dhruv.shah1409@gmail.com | linkedin.com/in/mayank-shah-b23969283 | github.com/Deathleio

Education

RCC Institute of Information Technology — Kolkata, West Bengal
B.Tech in Computer Science and Engineering (AIML); CGPA: 8.13/10.0 — *Aug. 2023 – Present*

Technical Skills

Core Languages: Python, SQL (MySQL, noSQL), Java, JavaScript
AI, ML & Vision: PyTorch, Scikit-Learn, YOLOv8, OpenCV, Fuzzy Logic (Mamdani), Pandas, NumPy
GenAI, RAG & NLP: LangChain, ChromaDB, Transformers (RoBERTa), LLM models , claude code , Tableu
Web & Frameworks: FastAPI, React.js,Tailwind CSS, RESTful APIs , POSTMAN API
Tools & Cloud: Docker, Git, GitHub, VS Code, Linux/Bash, Render, Vercel

Experience

Data Science & Engineering Intern — March 2026 – June 2026
Centre of Excellence on Data Science and Machine Learning — *Kolkata, India*

- Engineered the clinical transcription pipeline for the **AI Prescription Reader**, integrating **YOLOv8** for patient PII redaction and **Gemini Multimodal Vision** to structure EMR data and ICD-10 codes.
- Architected a **Dual-Layer Audit Engine** combining regex rules with **ChromaDB** formulary vector search; authored and submitted a research paper on the system for the **National Conference on e-Governance**.
- Collaborated across an 18-member engineering cohort to build automated validation scripts, data sanitization routines, and reporting dashboards for unstructured datasets.

Projects

AI Prescription Reader & Dual-Layer Audit Engine | *FastAPI, YOLOv8, ChromaDB, Gemini Vision, React.js* — March 2026 – June 2026

- Trained a custom **YOLOv8** model (**91.4% mAP@50**) to dynamically redact sensitive patient PII/signatures and utilized **Gemini Multimodal Vision** to transcribe handwritten regimens into ICD-10 diagnostic codes.
- Built a **Vector RAG pipeline** using **ChromaDB** indexing official CMS formularies to phonetically resolve noisy, handwritten drug shorthand with high retrieval accuracy.
- Architected a **Dual-Layer 100-Point Audit Engine** pairing deterministic rule validators with an LLM critic to detect transcript omissions and eliminate clinical hallucinations.

VeritasAI – Explainable Fake News Detection Platform | *Python, RoBERTa, PostgreSQL, FastAPI, Scikit-Learn*

- Engineered an NLP pipeline unifying 75K+ articles (WELFake, LIAR, CoAID) using a calibrated **Stacking Ensemble** and fine-tuned **RoBERTa** transformer for sub-second classification.
- Implemented an explainability module highlighting suspicious keywords and validating claims against the live **Wikipedia API**, backed by an indexed **PostgreSQL** query engine.
- Containerized and deployed a **FastAPI** microservice on Render with lazy model loading, serving real-time inferences to an interactive web dashboard on Vercel.

AURA – Glass-Box Multi-Agent Tutoring System | *FastAPI, LangGraph, ChromaDB, Gemini 2.5 Flash, React.js*

- Architected a stateful multi-agent tutoring graph via **LangGraph** and **Gemini 2.5 Flash**, dynamically routing learners between Socratic hinting, conceptual derivations, and direct remediation.
- Engineered a **Mamdani Fuzzy Inference System** scoring continuous accuracy, latency, and retries into 4 mastery tiers, grounded by an OpenStax **ChromaDB RAG** store with anti-leak sanitizers.
- Developed a full-stack glass-box UI in **React.js** displaying real-time agent cognitive node activations, state transitions, and timed diagnostic exam analytics.

Achievements & Extracurriculars

Research & Conferences: Represented department at National Conference on e-Governance; submitted research paper on clinical prescription auditing.

Academic Excellence: Maintained strong coursework standing with zero backlogs across all semesters.